



Mini-Capstone: PRD Assembly

Week 3 • Day 4 • Requirements Engineering & Mini-Capstone

TPM Academy — Non-AI keystone day



Day 4 Objectives

- Write **Section 8 Metrics & Validation** tied to your Tier Sheet and North Star (NS)
- Write **Section 9 Risks & Open Questions** — "no risks" is a fail
- Write **Section 10 Dependencies** with named owners
- Write **Section 11 Out-of-Scope Follow-ups** — the negotiation tool
- Run an **integration pass** — coherence + AI-prose detection — and lock the version for Friday



Today's Shape

Clock	Block	Outcome
09:00 – 09:15	Opening	Last day non-AI; Friday pairings posted
09:15 – 10:30	Block 1 + Activity 1	Section 8 Metrics & Validation
10:45 – 12:00	Block 2 + Activity 2	Section 9 Risks, Questions, Assumptions
13:00 – 14:15	Block 3 + Activity 3	Sections 10 + 11 Dependencies + Out-of-scope
14:30 – 16:00	Block 4 + Activity 4 + Wrap	Integration pass + lock

Block 1 — Section 8 Metrics & Validation

Reconnect the PRD to the strategy



Section 8 Anatomy

- **Primary metric** — one number, with target + defense
- **Counter-metric** — the guardrail from your NS Defense Card
- **Secondary metrics** (3 max) — directional indicators
- **Validation plan** — pre-launch / 7 days / 30 days / 90 days
- **Definition of "good enough"** — the success bar in plain words

Coherence check: the metrics here must match Week-2's Tier Sheet. If they don't, one of them is wrong.



Worked Section 8 (FieldPulse)

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## 8. Metrics & validation

### Primary metric
> Median reconcile-flow duration per dispatcher per shift
> Baseline: 45 min | Target: ≤ 12 min by day 30

### Counter-metric
> % of dispatchers calling support after using the new flow
> If this rises by > 3pp from baseline, we won the primary
> for the wrong reasons.

### Secondary metrics
- Reconcile-flow abandonment rate (declining)
- Manual override entries per dispatcher (declining)
- Mobile vs paper adoption ratio (mobile climbing)

### Validation plan
- Pre-launch: AC verification suite green
- Launch + 7 days: primary directionally correct
- Launch + 30 days: primary at or below 12 min
- Launch + 90 days: counter-metric within +3pp band
```

Activity 1 — Section 8 Metrics & Validation

Quads • 35 minutes

Pull primary, counter, secondaries from your Tier Sheet. Set targets with defenses. Write the validation plan.

- 10 min: pick metrics from Tier Sheet
- 15 min: set targets, write defenses
- 10 min: validation plan



10 + 15 + 10

Block 2 — Section 9 Risks, Questions, Assumptions

Honest naming of what could go wrong, what we don't know,
and what we're betting on



The Three Lists

Risks

Things that *could* go wrong. Each gets likelihood, impact, mitigation.

"No risks" is a fail.

Open questions

Things we don't know yet. Each gets an owner and a "needed by" date.

Assumptions

Things we're treating as true. Basis + revisit-if condition.

Watch for: "TBD" as owner. That's the unowned risk that slips. Force a name.



Worked Section 9 (excerpt)

Risks

Risk	L	I	Mitigation
Dispatchers refuse new flow because it's longer	M	H	Optional toggle in v1; observe adoption
Audit-log volume blows our log budget	L	M	Tier-2 sample for non-error events

Open questions

- Will operations VP accept temporary read-only manager view in week 1?
(Owner: Kim; needed by: 2026-05-12)
- Can audit store handle 24-month retention at this volume?
(Owner: Pat; needed by: 2026-05-19)

Assumptions

- Median dispatcher device is mid-tier Android on 4G LTE
(basis: device-fleet snapshot Q3; revisit if changes)
- 30-day adoption signal sufficient to declare success
(basis: prior reconcile rollout; revisit if behavior non-stationary)

Activity 2 — Section 9 Risks, Questions, Assumptions

Quads • 40 minutes

5 candidate risks → cull to 3–4 with mitigations. 3–5 open questions with owners. 2–4 assumptions with basis.

- 15 min: risks brainstorm + mitigations
- 10 min: open questions + owners
- 10 min: assumptions + basis
- 5 min: assign owners + by-when dates



15 + 10 + 10 + 5

Block 3 — Sections 10 + 11

Dependencies + Out-of-Scope

Make the network around your PRD explicit



Section 10 Dependencies

Dependency	Owner	What we need	By when	Status
Auth team — SSO scope	Lin	New scope `reconcile.write`	Sprint 1 Day 1	Confirmed
Data platform — schema	Pat	Schema review for 7 events	Sprint 1 Day 1	Pending
Mobile pipeline — signing	Kim	New signing cert	Pre-launch	Pending
CS — release messaging	Sam	Customer comms	T-7 days	Pending

Owner is a person, not a team. On the single sign-on (SSO) scope dependency, "the auth team" is unowned; "Lin from auth" is owned.

Section 11 Out-of-Scope Follow-ups

What this Product Requirements Document (PRD) acknowledges but won't ship in this iteration:

- **Tablet layout** — same flow, different breakpoint. Ticket: TICKET-1234.
- **Spanish localization** — copy + reading order. Ticket: TICKET-1235.
- **Multi-shop manager view** — cross-shop reconcile. Ticket: TICKET-1236.
- **Offline-first sync** — current ships offline draft + manual reconnect; full offline-first deferred. Ticket: TICKET-1237.

Why this matters: Section 11 is the negotiation tool for Week 6. "I see what you might also want; here's where it lives in the backlog."

Activity 3 — Sections 10 + 11

Quads • 40 minutes

Tabulate dependencies (with named owners). List out-of-scope follow-ups (specific, with ticket placeholders).

- 20 min: dependencies (owner / what / when / status)
- 15 min: out-of-scope follow-ups
- 5 min: status sanity check (most should be "Pending"; that's honest)



20 + 15 + 5

Block 4 — Integration Pass + Lock

Read top to bottom; fix incoherence; ship to Friday



The Integration Checklist

Read the PRD top to bottom: do the acceptance criteria (AC) and non-functional requirements (NFR) still cohere?

Check	Pass criterion
Section 1 → Section 2 flow	Reader is motivated to keep reading
Section 3 goals tied to Section 8 metrics	Each goal references a metric
Section 5 sketch ↔ Section 6 AC	Each happy AC corresponds to a sketch step
Section 6 AC vs Section 7 NFRs	NFRs cover what AC don't
Section 7 NFRs ↔ Section 8 observability	Every metric has an obs NFR
Section 4 scope-out → Section 11 follow-ups	Items left out appear in follow-ups
Section 9 risks named, owned, mitigated	No "no risks"
No AI-generic prose	Voice is consistent quad voice

✂ AI-Generic Prose Detector

Tells to eliminate:

- "It is important to note that..."
- "By leveraging X, we can unlock Y..."
- "This robust solution will empower stakeholders to..."
- "In today's fast-paced environment..."
- Three adjectives in a row when one would do
- Sentences that sound true but say nothing specific

The test: if a section could appear in any PRD for any product, rewrite it for *your* product.

Activity 4 — Integration Pass + Lock

Quads • 45 minutes • Wrap

Solo read-through, pool issues, fix in priority order, lock. Save as "v0 — for Friday."

- 15 min: solo read-through with margin notes
- 10 min: pool + de-dupe issues
- 15 min: fix coherence first, prose second
- 5 min: lock (Status: In review) + save v0



15 + 10 + 15 + 5 · 15-min wrap-share



Day 4 Wrap

What we shipped

- Sections 1–11 complete PRD
- Tier-Sheet linked metrics
- Risks owned and mitigated
- Dependencies named
- Integration pass + lock

What opens tomorrow

- **PRD Review day**
- Morning: primary review
- Afternoon: revisions + secondary review
- End of day: signed-off PRD

Bring to Day 5: Your locked v0 PRD. Friday cadence is different — see the morning brief.

Questions & Reflection

Where in your PRD will tomorrow's reviewer push hardest?